

Introduction

Multi-omics experiments measure transcriptome, methylome, proteome, etc. on the same samples. Joint analysis exposes coordinated variation across layers that single-layer analyses miss – cis-regulatory effects of methylation, mRNA-protein correspondence, or integrated disease subtypes.

Prerequisites

Single-omic analysis (DE, methylation, proteomics); PCA / factor analysis basics.

Theory

MOFA (Multi-Omics Factor Analysis) learns latent factors explaining variation across omics layers. Each factor has per-layer weights: factors loading strongly on multiple layers capture shared signal; single-layer factors reflect layer-specific variation. MOFA2 adds sparsity for interpretability.

Similarity network fusion (SNF) and **iCluster+** offer alternative approaches; **DIABLO (mixOmics)** builds supervised multi-omic classifiers.

Assumptions

Samples are matched across layers; layer-specific noise and scaling are handled by the model; sufficient sample size for factor inference (typically > 50).

R Implementation

```
library(MOFA2)

set.seed(2026)
n <- 50
# Latent factor drives correlated variation across 2 layers
z <- rnorm(n)

# Layer 1: 200 features (e.g., RNA)
m_rna <- matrix(rnorm(200 * n, 0, 1), 200, n)
m_rna[1:20, ] <- m_rna[1:20, ] + outer(rep(1.5, 20), z)

# Layer 2: 100 features (e.g., methylation)
m_meth <- matrix(rnorm(100 * n, 0, 1), 100, n)
m_meth[1:10, ] <- m_meth[1:10, ] - outer(rep(1.2, 10), z)

colnames(m_rna) <- colnames(m_meth) <- paste0("S", 1:n)
rownames(m_rna) <- paste0("rna_", 1:200)
rownames(m_meth) <- paste0("meth_", 1:100)

mofa <- create_mofa(list(RNA = m_rna, Methylation = m_meth))
# Illustrative: actual fitting requires Python backend (reticulate + mofapy2).
# model_opts <- get_default_model_options(mofa)
# model_opts$num_factors <- 3
# mofa_trained <- run_mofa(mofa, training_data_options = list(),
#                           model_options = model_opts, outfile = tempfile())

mofa
```

Output & Results

A trained MOFA model with latent-factor scores per sample, per-layer feature weights, and variance decomposition per factor x layer.

Interpretation

“MOFA identified 5 latent factors; factor 1 explained 32 % of RNA and 18 % of methylation variance, linked to cancer subtype; factor 2 was RNA-specific and reflected proliferation signature.”

Practical Tips

- Standardise each layer before MOFA (mean = 0, SD = 1) so no one layer dominates.
- Layer-specific factors often reflect technical artefacts; use them to diagnose batch effects.
- DIABLO in `mixOmics` is the better choice when integration is supervised by an outcome.
- MOFA requires Python via `reticulate`; test the install with `run_mofa2_docs()` before large runs.
- Interpret factor weights with domain knowledge; algorithmic factors without biology are uninformative.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat